Lookahead Bias in Large Language Models: Measurement and Mitigation Techniques

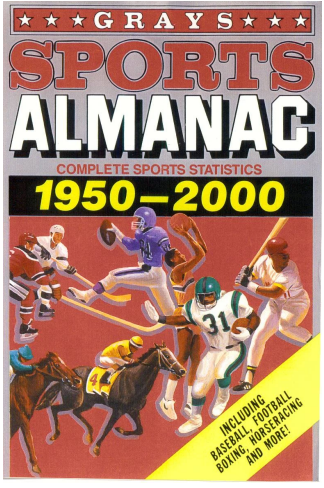
Asaf Manela
Washington University in St. Louis

with Joseph Engelberg, Songrun He, Linying Lv, William Mullins, Luka Vulicevic, Jimmy Wu

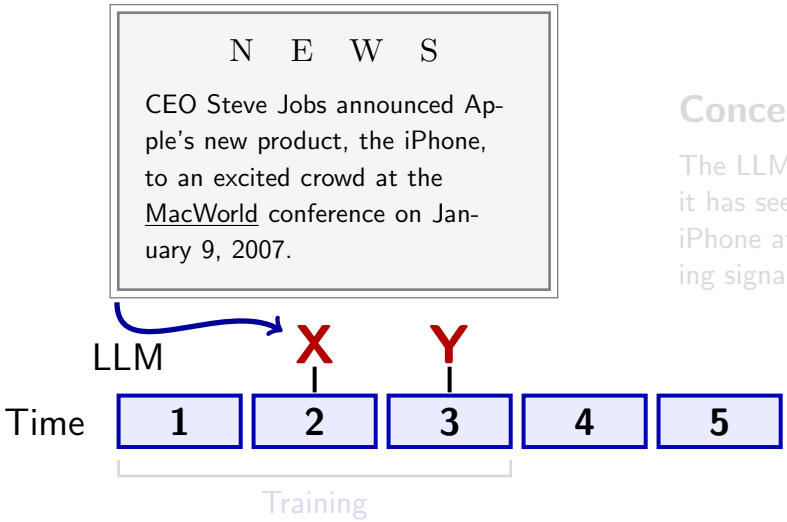
May 2026

The problem: Lookahead bias in LLMs

- Large language models (LLMs) now permeate social sciences
- They let us test hypotheses previously unquantifiable
- But they are trained on data that did not exist at the historical moment
- Lookahead bias and training leakage raise doubts about LLM-based empirical findings
  - Glasserman and Lin (2023), Sarkar and Vafa (2024), Ludwig, Mullainathan, and Rambachan (2024), Levy (2024), Lopez-Lira, Tang, and Zhu (2025)
- Finance is particularly sensitive—market efficiency requires prices reflect only facts known at the time



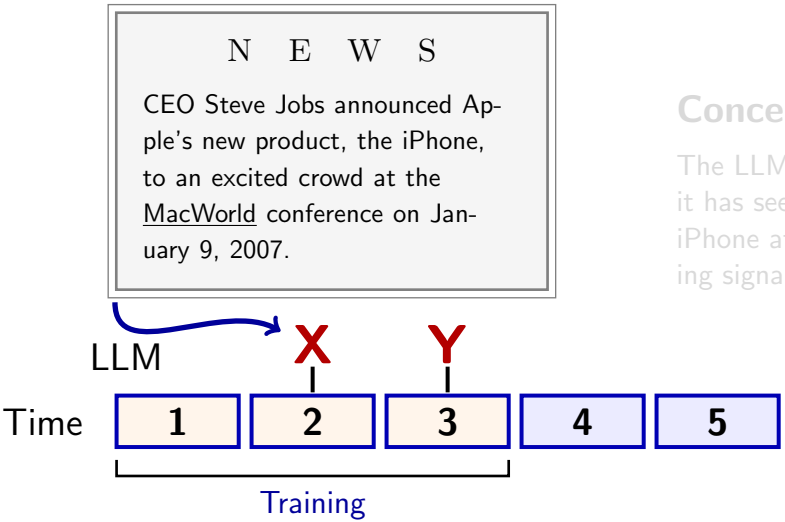
Stylized example: Extract signal X to predict Y



Concern

The LLM will use information it has seen about Apple or the iPhone at *Time 3* when producing signal **X** from the text

Stylized example: Extract signal X to predict Y



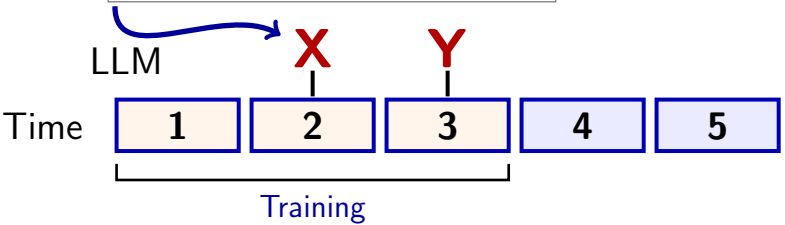
Concern

The LLM will use information it has seen about Apple or the iPhone at *Time 3* when producing signal **X** from the text

Stylized example: Extract signal X to predict Y

N E W S

CEO Steve Jobs announced Apple's new product, the iPhone, to an excited crowd at the MacWorld conference on January 9, 2007.



Concern

The LLM will use information it has seen about Apple or the iPhone at *Time 3* when producing signal **X** from the text

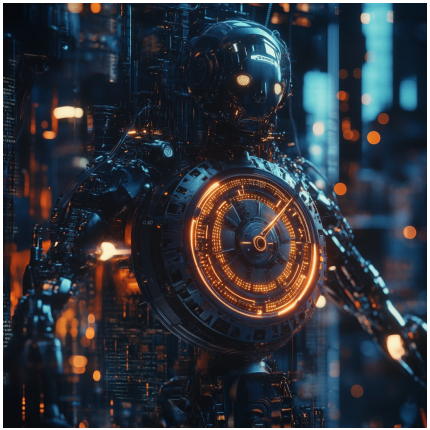
Agenda

1. **Chronologically Consistent Large Language Models** (He, Lv, Manela, Wu)
   - Train LLMs exclusively on data available before each time period
2. **Instruction Tuning Chronologically Consistent LLMs** (He, Lv, Manela, Wu)
   - Extend chronological consistency to instruction-tuned chat models
3. **Entity Neutering** (Engelberg, Manela, Mullins, Vulicevic)
   - Agentic masking approach to prevent LLM from inferring firm identity or timing

Paper 1: Chronologically Consistent Large Language Models

Songrun He, Linying Lv, Asaf Manela, Jimmy Wu

ChronoLLMs: What we do



- We train chronologically consistent LLMs exclusively on preceding text
  - $ChronoBERT_{1999}, \ldots, ChronoBERT_{2024}$
  - $ChronoGPT_{1999}, \ldots, ChronoGPT_{2024}$
  - Available on Hugging Face
- Simple, right?
- Ensuring these models are competitive with prominent, high-performing counterparts is hard!

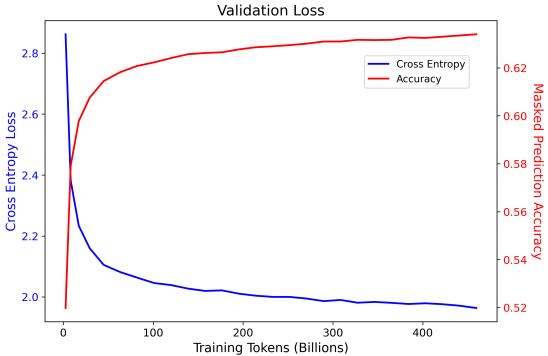
Pretraining chronologically consistent LLMs

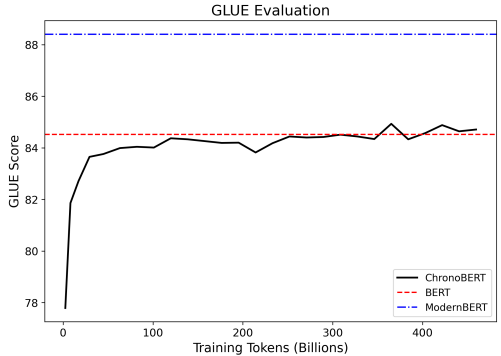
- Training LLMs is usually split into:
  1. Pretraining to predict missing words in text sequences
  2. Finetuning for specific applications (e.g. chat, Q&A, reasoning)
- We pretrain $\mathit{ChronoBERT}_t$ and $\mathit{ChronoGPT}_t$ only on text available before t
  - For example, web pages crawled in 2005 would be used for pretraining $\mathit{ChronoBERT}_{2005}$ using $\mathit{ChronoBERT}_{2004}$ as a starting point

Pretraining chronologically consistent LLMs

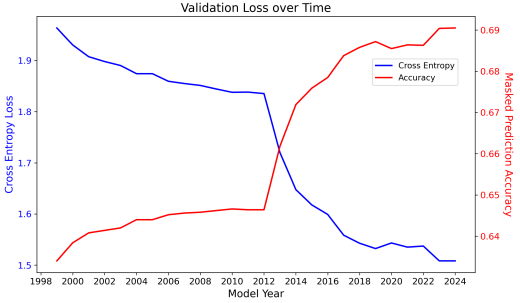
- Ensuring these models are competitive with larger counterparts poses two challenges:
  1. Limited compute
  2. Limited historical data
- We draw on efficient training methods (Portes et al 2023; Warner et al 2024; Jordan et al 2024) to lower computing costs
- Follow Gunasekar et al. (2023) by selecting diverse, high-quality data, carefully filtered by publication date to maximize information gained from a limited corpus
- Follow Muennighoff et al (2023) insights to train over multiple epochs to maximize learning from the available corpus
- Initial 1999 models are trained on a corpus of 7 billion tokens over multiple epochs
- Incremental training from 2000 to 2024 on a corpus of 112 billion tokens

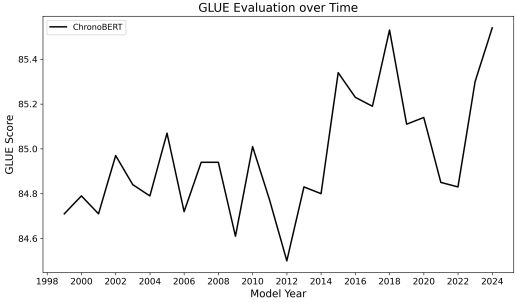
ChronoBERT$_{1999}$ surpasses BERT on GLUE benchmarks



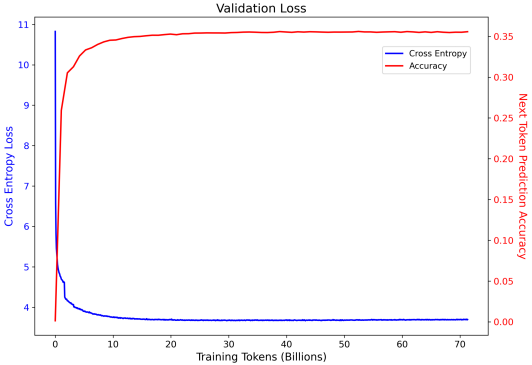


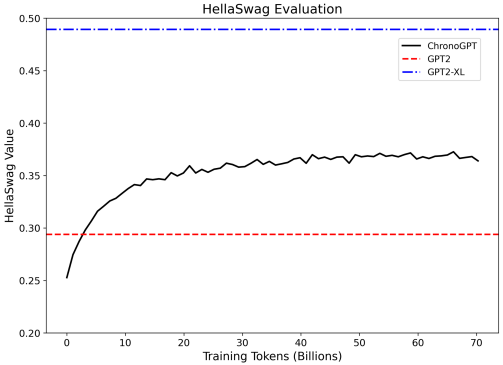
Subsequent ChronoBERT models improve over time



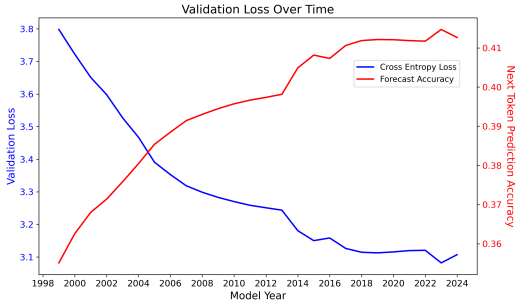


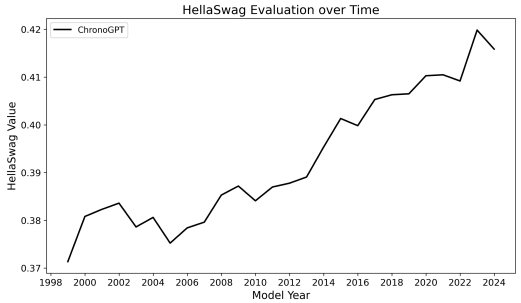
ChronoGPT$_{1999}$ improves to the point it surpasses GPT-2 on HellaSwag token generation





Subsequent ChronoGPT models improve over time





ChronoBERT and ChronoGPT exhibit superior language understanding relative to similar-sized models and comparable to much larger Llama

GLUE score evaluation of language understanding

	Parameters	Context Tokens	Knowledge Cutoff	GLUE
ChronoBERT_{1999}	149M	1,024	December, 1999	84.71
⋮	⋮	⋮	⋮	
ChronoBERT_{2024}	149M	1,024	December, 2024	85.54
ChronoGPT_{1999}	1,552M	1,792	December, 1999	77.42
⋮	⋮	⋮	⋮	
ChronoGPT_{2024}	1,552M	1,792	December, 2024	81.63
BERT	110M	512	October, 2018	84.52
FinBERT	110M	512	December, 2019	76.41
StoriesLM	110M	512	December, 1963	79.36
Llama 3.1	8,030M	128,000	December 2023	86.31

ChronoBERT and ChronoGPT are comparable to Llama in predicting next-day stock returns with news

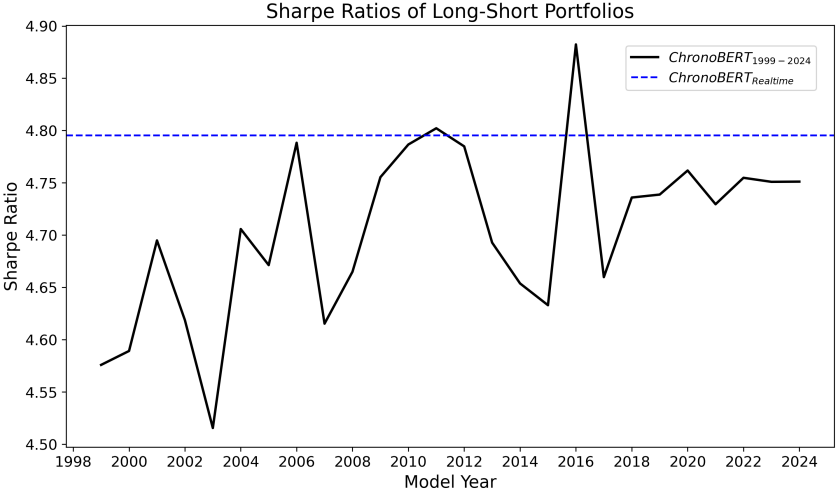
Performance of LLM portfolios

	$\text{ChronoBERT}_{\text{Realtime}}$			$\text{ChronoGPT}_{\text{Realtime}}$			Llama 3.1			GPT-2 XL		
	Mean	SD	SR	Mean	SD	SR	Mean	SD	SR	Mean	SD	SR
Low (L)	-23.30	25.86	-0.90	-19.82	25.15	-0.79	-23.71	26.15	-0.91	-22.25	25.75	-0.86
5	3.94	24.22	0.16	7.18	24.54	0.29	7.47	24.65	0.30	6.24	24.30	0.26
High (H)	37.71	24.53	1.54	43.08	25.41	1.70	42.20	25.05	1.68	38.25	24.53	1.56
H-L	**61.02**	**12.72**	**4.80**	**62.90**	**12.78**	**4.92**	**65.91**	**13.46**	**4.90**	**60.51**	**13.24**	**4.57**

- ChronoBERT (SR 4.80) and ChronoGPT (SR 4.92) comparable to much larger Llama (SR 4.90)
- Find modest lookahead bias in this setting

Time-aligned LLMs predict better

Portfolios performance across ChronoBERT vintages



ChronoLLMs: Takeaways

- Chronological inconsistency can bias LLM-based empirical estimates
- We train a suite of chronologically consistent LLMs
- ChronoBERT and ChronoGPT exhibit superior language understanding relative to similar-sized models
- Performance comparable to much larger (and inconsistent) Llama
- Find modest lookahead bias in a news-return prediction application
- Models available on Hugging Face for researchers to evaluate bias in their applications
- If you need embeddings ChronoBERT and ChronoGPT are your workhorses

Paper 2: Instruction Tuning Chronologically Consistent Language Models

Songrun He, Linying Lv, Asaf Manela, Jimmy Wu

Instruction tuning and lookahead bias

- ChronoBERT and ChronoGPT are pretrained but not finetuned for chat
- These are completion models, so if you expect ChatGPT-like behavior you will be disappointed
- LLMs are finetuned for chat as a second step to get Generative AI behavior
- This step is usually done with considerably less data than the pretraining
- But using off-the-shelf datasets for chat can introduce lookahead bias

Instruction tuning can introduce lookahead bias

Example: training a 2007 model on a 2009 instruction dataset

User:
What are the risks of investing in mortgage-backed securities?

Assistant:
Mortgage-backed securities carry several risks:
1. Credit risk: As seen in the 2008 subprime mortgage crisis, widespread defaults can cause catastrophic losses
2. Prepayment risk: Borrowers may refinance when rates fall
3. Interest rate risk: Values decline when rates rise
The collapse of Lehman Brothers demonstrated how interconnected these risks can be...

Instruction tuning chronologically consistent language models

- We train a family of ChronoGPT-Instruct models that are finetuned for chat
- We curate a dataset of 425,000 conversations between a user and an assistant
- Contain only knowledge available before 2000
- ChronoGPT-Instruct models and the instruction-tuning dataset we curated are available on Hugging Face



LLM-based filtering prompt for chronological consistency

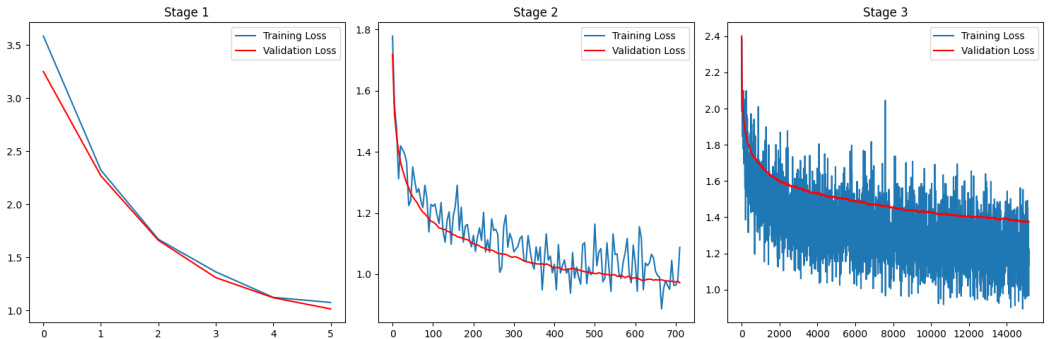
Prompt: You are provided with a user-assistant interaction. Your task is to determine whether the conversation contains any information that would have been unavailable or irrelevant prior to the year 2000.
Specifically, indicate whether the message includes any direct or indirect reference to:
1. A concept, company, product, technology, event, online review, or terminology that was created, discovered, or publicly introduced after 1999, or
2. A subject that only gained significant economic, cultural, scientific, or technological relevance after 1999, even if it existed before that date.
If such a reference is present anywhere in the conversation, return: 1
If the conversation is entirely composed of content that could have been generated using only knowledge available prior to 2000, return: 0
Clarifications:
- For conversations evaluated as low quality, also assign a label of 1.
- In cases of uncertainty or ambiguity, adopt a conservative approach and assign a label of 1.
- References to post-1999 entities such as GPT models, Kubernetes, TikTok, blockchain, COVID-19, Tesla, or similar modern constructs are strong indicators of a label of 1.
Return your answer strictly as a JSON object with the following fields:
- "label": either 0 or 1
- "confidence": a number from 0 to 10 (higher means more certain)
- "suspected term": a brief phrase (1-3 words) that triggered your label decision, or "none" if label=0
Example output: {"label": 1, "confidence": 9, "suspected term": "GPT-3"}
Here is the message: {conversation}

Instruction tuning datasets

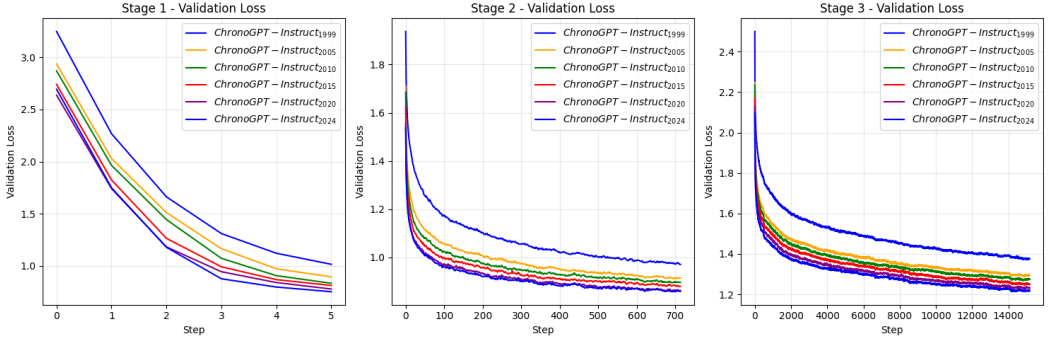
Stage	SFT data	# examples	Avg. conv. length
1	LLMs-from-scratch	1,097	102
2	GPT-3 self-generated	67,136	183
3	Tulu-3 SFT mixture	356,886	2,513

- Curriculum grows in cognitive load and sequence length
- Filter to exclude non-English records and code snippets
- Retain only entries labeled 0 ("pre-2000 knowledge") with maximum confidence

Instruction tuning ChronoGPT-1999



Systematic improvements from earlier to later vintages



Validation: ChronoGPT-Instruct knows only what it should

U.S. Presidents in chronological order:
Took office in $\{year_{p-3}+1\}$*: President* $\{name_{p-3}\}$
Took office in $\{year_{p-2}+1\}$*: President* $\{name_{p-2}\}$
Took office in $\{year_{p-1}+1\}$*: President* $\{name_{p-1}\}$
Took office in $\{year_{p}+1\}$*: President* _____

							Accuracy	
Election year:	1992	2000	2008	2016	2020	2024	Pre	Post
Correct output	Bill Clinton	George W	Barack Obama	Donald Trump	Joe Biden	Donald Trump		
GPT-2	**Bill Clinton**	Bill Clinton	**Barack Obama**	**Donald Trump**	George W	George W	3/4	0/2
GPT-2 XL	**Bill Clinton**	**George W**	**Barack Obama**	**Donald Trump**	James A	James Mattis	4/4	0/2
Llama-3.2-3B-Instruct	**Bill Clinton**	**George W**	**Barack Obama**	**Donald Trump**	**Joe Biden**	R.	5/5	0/1
Qwen-1.5-1.8B-Chat	**Bill Clinton**	**George W**	**Barack Obama**	**Donald Trump**	**Joe Biden**	Kamala	5/5	0/1
ChronoGPT-Instruct$_{\text{Realtime}}$	—	Bill Clinton	George W	John F	elect2019	Joe Biden	—	0/5
ChronoGPT-Instruct$_{2000}$	**Bill Clinton**	Bill Clinton	Bill Clinton	Bill Clinton	Bill Clinton	Bill Clinton	1/2	0/4
ChronoGPT-Instruct$_{2008}$	**Bill Clinton**	**George W**	George W	George W	George W	George W	2/3	0/3
ChronoGPT-Instruct$_{2012}$	**Bill Clinton**	**George W**	**Barack Obama**	Barack Obama	Bill Gates	George W	3/3	0/3
ChronoGPT-Instruct$_{2016}$	**Bill Clinton**	**George W**	George W	Martin Luther King	George W	George W	2/4	0/2
ChronoGPT-Instruct$_{2020}$	**Bill Clinton**	**George W**	George W	**Donald Trump**	Bill Clinton	George W	3/5	0/1
ChronoGPT-Instruct$_{2024}$	**Bill Clinton**	**George W**	George W	**Donald Trump**	**Joe Biden**	Joe Biden	4/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							67/83	0/73

blue = correct prediction
gray area = post-knowledge cutoff prediction

Validation: ChronoGPT-Instruct knows only what it should

Next-token predictions of event-defining tokens

Event year:	2001	2003	2008	2016	2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
GPT-2 XL	**scandal**, in	**SARS**,	**crisis**. The	. The UK	Great Recession."	Alexa,"	3/4	0/2
Llama-3.2-3B-Instruct	**scandal**, which	**SARS** (	**crisis**, which	**referendum**. The	**COVID**-19	LLaMA	5/6	—
Qwen-1.5-1.8B-Chat	**scandal**, which	**SARS**.	**crisis**. The	**referendum**. The	**COVID**-1	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
ChronoGPT-Instruct$_{2002}$	**scandal**. It	the "H	market, which	. The market	Asian flu."	AI-1	1/1	0/5
ChronoGPT-Instruct$_{2004}$	**scandal**. It	**SARS**,	market, which	hip.	Asian flu."	Chatbot".	2/2	0/4
ChronoGPT-Instruct$_{2008}$	**scandal**. It	the **SARS**	**crisis**, which	.	Asian flu."	Chatbot".	3/3	0/3
ChronoGPT-Instruct$_{2012}$	**scandal**, which	the **SARS**	**crisis**, which	vote.	Asian flu"	The Language Model	3/3	0/3
ChronoGPT-Instruct$_{2016}$	**scandal**. It	the **SARS**	**crisis**, which	. The European	Asian flu"	The Chatbot	3/4	0/2
ChronoGPT-Instruct$_{2020}$	**scandal**, which	**SARS**-	**crisis**, which	**referendum**. The	Spanish flu"	AI Language Model	4/5	0/1
ChronoGPT-Instruct$_{2022}$	Corporation collapse,	the **SARS**	**crisis**, which	**referendum**. The	**coronav**	AI Assistant"	5/6	—
ChronoGPT-Instruct$_{2024}$	**scandal**. It	**SARS**,	**crisis**, which	**referendum**. The	**COVID**-	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

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Validation: ChronoGPT-Instruct knows only what it should

The Sarbanes-Oxley Act was introduced in response to the 2001 Enron _____

Event year:	▼ 2001	2003	2008	2016	2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
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Validation: ChronoGPT-Instruct knows only what it should

In 2003, a major public health crisis was the outbreak of the virus known as _____

Event year:	2001	▼ 2003	2008	2016	2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
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ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
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ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

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Validation: ChronoGPT-Instruct knows only what it should

In 2008, the global economy was dominated by the subprime mortgage _____

Event year:	2001	2003	▼ 2008	2016	2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
GPT-2 XL	**scandal**, in	**SARS**,	**crisis**. The	. The UK	Great Recession."	Alexa,"	3/4	0/2
Llama-3.2-3B-Instruct	**scandal**, which	**SARS** (	**crisis**, which	**referendum**. The	**COVID**-19	LLaMA	5/6	—
Qwen-1.5-1.8B-Chat	**scandal**, which	**SARS**.	**crisis**. The	**referendum**. The	**COVID**-1	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
ChronoGPT-Instruct$_{2002}$	**scandal**. It	the "H	market, which	. The market	Asian flu."	AI-1	1/1	0/5
ChronoGPT-Instruct$_{2004}$	**scandal**. It	**SARS**,	market, which	hip.	Asian flu."	Chatbot".	2/2	0/4
ChronoGPT-Instruct$_{2008}$	**scandal**. It	the **SARS**	**crisis**, which	.	Asian flu."	Chatbot".	3/3	0/3
ChronoGPT-Instruct$_{2012}$	**scandal**, which	the **SARS**	**crisis**, which	vote.	Asian flu"	The Language Model	3/3	0/3
ChronoGPT-Instruct$_{2016}$	**scandal**. It	the **SARS**	**crisis**, which	. The European	Asian flu"	The Chatbot	3/4	0/2
ChronoGPT-Instruct$_{2020}$	**scandal**, which	**SARS**-	**crisis**, which	**referendum**. The	Spanish flu"	AI Language Model	4/5	0/1
ChronoGPT-Instruct$_{2022}$	Corporation collapse,	the **SARS**	**crisis**, which	**referendum**. The	**coronav**	AI Assistant"	5/6	—
ChronoGPT-Instruct$_{2024}$	**scandal**. It	**SARS**,	**crisis**, which	**referendum**. The	**COVID**-	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

blue = correct prediction

gray area = post-knowledge cutoff prediction

Validation: ChronoGPT-Instruct knows only what it should

In 2016, market volatility increased surrounding the general vote known as the Brexit

Event year:	2001	2003	2008	▼ 2016	2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
GPT-2 XL	**scandal**, in	**SARS**,	**crisis**. The	. The UK	Great Recession."	Alexa,"	3/4	0/2
Llama-3.2-3B-Instruct	**scandal**, which	**SARS** (	**crisis**, which	**referendum**. The	**COVID**-19	LLaMA	5/6	—
Qwen-1.5-1.8B-Chat	**scandal**, which	**SARS**.	**crisis**. The	**referendum**. The	**COVID**-1	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
ChronoGPT-Instruct$_{2002}$	**scandal**. It	the "H	market, which	. The market	Asian flu."	AI-1	1/1	0/5
ChronoGPT-Instruct$_{2004}$	**scandal**. It	**SARS**,	market, which	hip.	Asian flu."	Chatbot".	2/2	0/4
ChronoGPT-Instruct$_{2008}$	**scandal**. It	the **SARS**	**crisis**, which	.	Asian flu."	Chatbot".	3/3	0/3
ChronoGPT-Instruct$_{2012}$	**scandal**, which	the **SARS**	**crisis**, which	vote.	Asian flu"	The Language Model	3/3	0/3
ChronoGPT-Instruct$_{2016}$	**scandal**. It	the **SARS**	**crisis**, which	. The European	Asian flu"	The Chatbot	3/4	0/2
ChronoGPT-Instruct$_{2020}$	**scandal**, which	**SARS**-	**crisis**, which	**referendum**. The	Spanish flu"	AI Language Model	4/5	0/1
ChronoGPT-Instruct$_{2022}$	Corporation collapse,	the **SARS**	**crisis**, which	**referendum**. The	**coronav**	AI Assistant"	5/6	—
ChronoGPT-Instruct$_{2024}$	**scandal**. It	**SARS**,	**crisis**, which	**referendum**. The	**COVID**-	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

blue = correct prediction

gray area = post-knowledge cutoff prediction

Validation: ChronoGPT-Instruct knows only what it should

In 2020, the global economy was devastated by the health crisis known as the "_____

Event year:	2001	2003	2008	2016	▼ 2020	2022	Accuracy Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
GPT-2 XL	**scandal**, in	**SARS**,	**crisis**. The	. The UK	Great Recession."	Alexa,"	3/4	0/2
Llama-3.2-3B-Instruct	**scandal**, which	**SARS** (	**crisis**, which	**referendum**. The	**COVID**-19	LLaMA	5/6	—
Qwen-1.5-1.8B-Chat	**scandal**, which	**SARS**.	**crisis**. The	**referendum**. The	**COVID**-1	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
ChronoGPT-Instruct$_{2002}$	**scandal**. It	the "H	market, which	. The market	Asian flu."	AI-1	1/1	0/5
ChronoGPT-Instruct$_{2004}$	**scandal**. It	**SARS**,	market, which	hip.	Asian flu."	Chatbot".	2/2	0/4
ChronoGPT-Instruct$_{2008}$	**scandal**. It	the **SARS**	**crisis**, which	.	Asian flu."	Chatbot".	3/3	0/3
ChronoGPT-Instruct$_{2012}$	**scandal**, which	the **SARS**	**crisis**, which	vote.	Asian flu"	The Language Model	3/3	0/3
ChronoGPT-Instruct$_{2016}$	**scandal**. It	the **SARS**	**crisis**, which	. The European	Asian flu"	The Chatbot	3/4	0/2
ChronoGPT-Instruct$_{2020}$	**scandal**, which	**SARS**-	**crisis**, which	**referendum**. The	Spanish flu"	AI Language Model	4/5	0/1
ChronoGPT-Instruct$_{2022}$	Corporation collapse,	the **SARS**	**crisis**, which	**referendum**. The	**coronav**	AI Assistant"	5/6	—
ChronoGPT-Instruct$_{2024}$	**scandal**. It	**SARS**,	**crisis**, which	**referendum**. The	**COVID**-	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

blue = correct prediction

gray area = post-knowledge cutoff prediction

Validation: ChronoGPT-Instruct knows only what it should

In 2022, a major milestone for generative AI was marked by the release of the AI chatbot known as "_____

						▼	Accuracy	
Event year:	2001	2003	2008	2016	2020	2022	Pre	Post
Correct output	scandal	SARS	crisis	referendum	COVID/coronavirus	ChatGPT		
GPT-2	**scandal**.\n	chikung	**crisis**, which	vote, with	Great Recession."	The AI Bot	2/4	0/2
GPT-2 XL	**scandal**, in	**SARS**,	**crisis**. The	. The UK	Great Recession."	Alexa,"	3/4	0/2
Llama-3.2-3B-Instruct	**scandal**, which	**SARS** (	**crisis**, which	**referendum**. The	**COVID**-19	LLaMA	5/6	—
Qwen-1.5-1.8B-Chat	**scandal**, which	**SARS**.	**crisis**. The	**referendum**. The	**COVID**-1	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{\text{Realtime}}$	ment Act,	the "H	market, which	-Elli	H1N	AI Assistant"	—	0/6
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ChronoGPT-Instruct$_{2004}$	**scandal**. It	**SARS**,	market, which	hip.	Asian flu."	Chatbot".	2/2	0/4
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ChronoGPT-Instruct$_{2024}$	**scandal**. It	**SARS**,	**crisis**, which	**referendum**. The	**COVID**-	**ChatGPT**	6/6	—
ChronoGPT-Instruct$_{1999}$ through ChronoGPT-Instruct$_{2024}$							76/80	0/76

blue = correct prediction

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Application: Prompt-based trading portfolios

- Lopez-Lira and Tang (2023) demonstrate that prompting LLMs can yield robust trading signals
- We follow their approach using Dow Jones Newswire headlines from 2007 to 2023

```
### Instruction:
Classify this news headline as either FAVORABLE, or UNFAVORABLE,
or UNCLEAR for the stock price of {company}.

### Input:
{headlines}

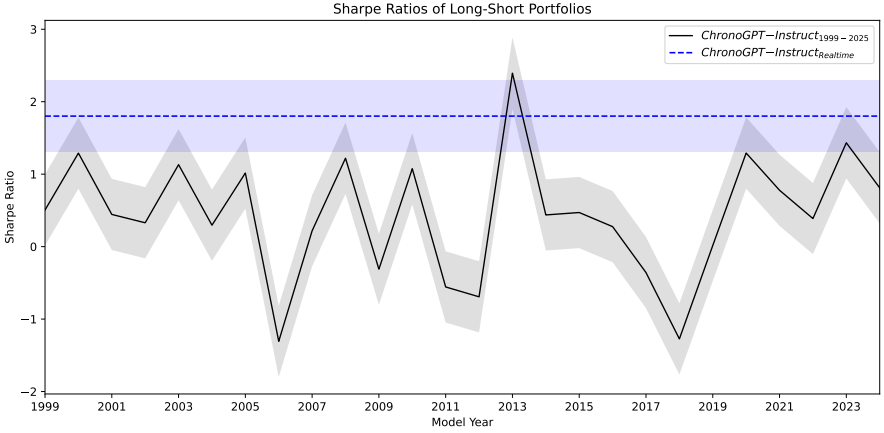
### Response:
```

ChronoGPT performance is comparable to similar-sized Qwen and Llama

But ChronoGPT is the only chronologically consistent model

	ChronoGPT-1.5B-Instruct			Qwen-1.5-1.8B-Chat		
	Mean	SD	SR	Mean	SD	SR
Unfavorable (L)	1.35	24.53	0.05	-1.67	25.90	-0.06
Unclear	0.14	29.03	0.00	7.20	78.95	0.09
Favorable (H)	9.51	23.33	0.41	10.55	22.72	0.46
$H - L$	8.17	8.63	0.95	12.21	8.00	1.53
	Llama-3.2-3B-Instruct			Llama-3.2-1B-Instruct		
	Mean	SD	SR	Mean	SD	SR
Unfavorable (L)	-1.11	25.68	-0.04	4.67	23.64	0.20
Unclear	6.71	23.17	0.29	14.01	23.04	0.61
Favorable (H)	13.46	23.31	0.58	7.31	23.91	0.31
$H - L$	14.58	8.31	1.76	2.64	6.91	0.38

Portfolio performance across ChronoGPT-Instruct vintages



- Sharpe ratios across vintages corroborate the envelope pattern of He et al. (2025)
- Modest lookahead bias: latest vintage is not the top performer in every period

ChronoGPT-Instruct: Takeaways

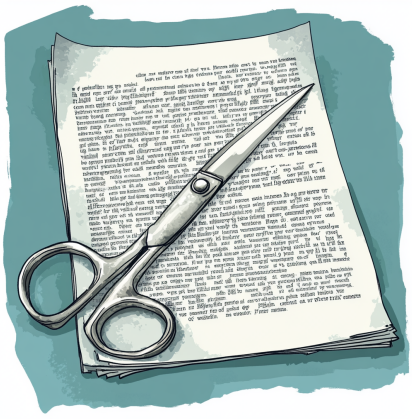
- Off-the-shelf instruction datasets can leak future knowledge even into a chronologically consistent base model
- We curate a pre-2000 instruction-tuning dataset and train a suite of chronologically consistent chat models: ChronoGPT-Instruct$_{1999}$, . . . , ChronoGPT-Instruct$_{2024}$
- If you need a chat-capable model with no lookahead bias, ChronoGPT-Instruct is your workhorse
- Models and dataset on Hugging Face

Paper 3: Entity Neutering

Joseph Engelberg, Asaf Manela, William Mullins, Luka Vulicevic

Entity neutering: Key idea

- Entity Neutering: ask the LLM to use its judgement in removing all identifying information
- Specifically, instruct the LLM to mask and paraphrase iteratively until the subject firm is unrecognizable by the LLM
- Prevents inference of firm identity or timing
- Maintains semantic content for sentiment extraction

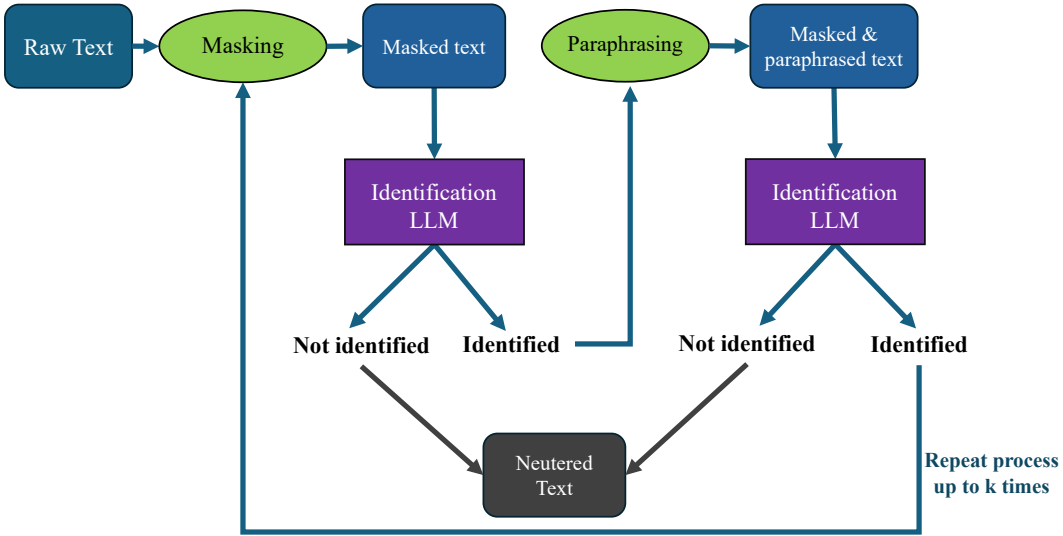


An example

Original Text	Firm Masked	Entity Neutered
Coca-Cola (KO) hasn't been a popular stock of late because of weak **soda** sales (volume logged its first quarterly drop in **15 years** in **1Q**). But **RBC** says forthcoming **2Q** results could be the strongest in a while amid a **later Easter** versus **2013**, a **World Cup** boost and the **"Share A Coke"** campaign in which consumers personalize **Coke cans** with names. **RBC** is forecasting **4.4% volume growth**, including **3%** in the **US**, up from **2%** globally in **1Q** and **1%** in **4Q**.	*Entity_A (Ticker_A)* hasn't been a popular stock of late because of weak **soda** sales (volume logged its first quarterly drop in **15 years** in **1Q**). But **RBC** says forthcoming **2Q** results could be the strongest in a while amid a **later Easter** versus **2013**, a **World Cup** boost and the "**Share A** *Entity_A*" campaign in which consumers personalize *Entity_A* **cans** with names. **RBC** is forecasting **4.4% volume growth**, including **3%** in the **US**, up from **2%** globally in **1Q** and **1%** in **4Q**.	*Entity_A's* most recent performance has failed to gain momentum due to soft demand for *sku_alpha*, having their first fall in demand in *a_years*. Nevertheless, *Entity_B* indicates the upcoming *period_z* could surpass the *previous_time_performance*, helped by an *annual_event*, a *marketing_promo* uptick, and a program allowing users to customize *sku_beta* with a *tag_id*. *Entity_B* expects *x_volume_growth*, including pockets of growth across *region_x* and *region_y*.

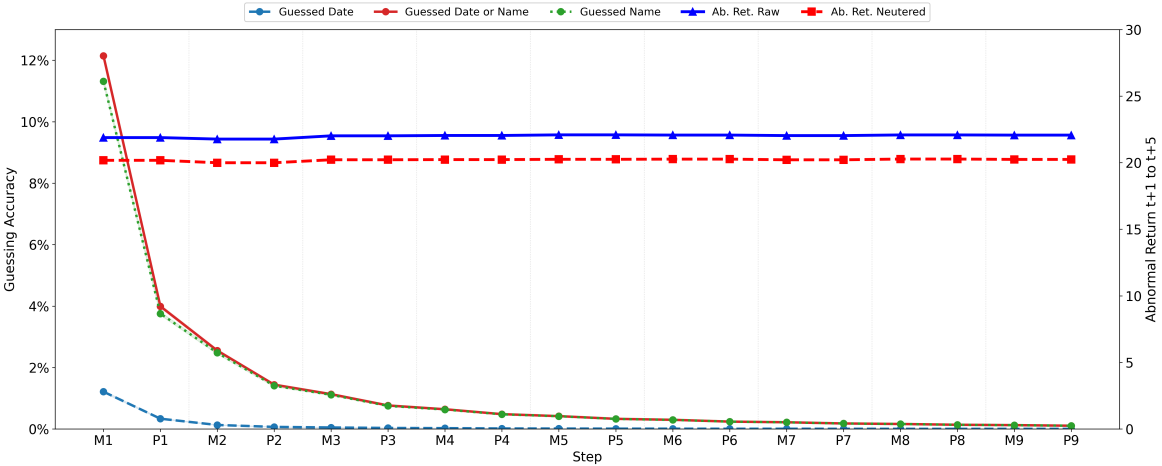
Source: Dow Jones Newswire Service, July 10, 2014

Agentic approach to neutering



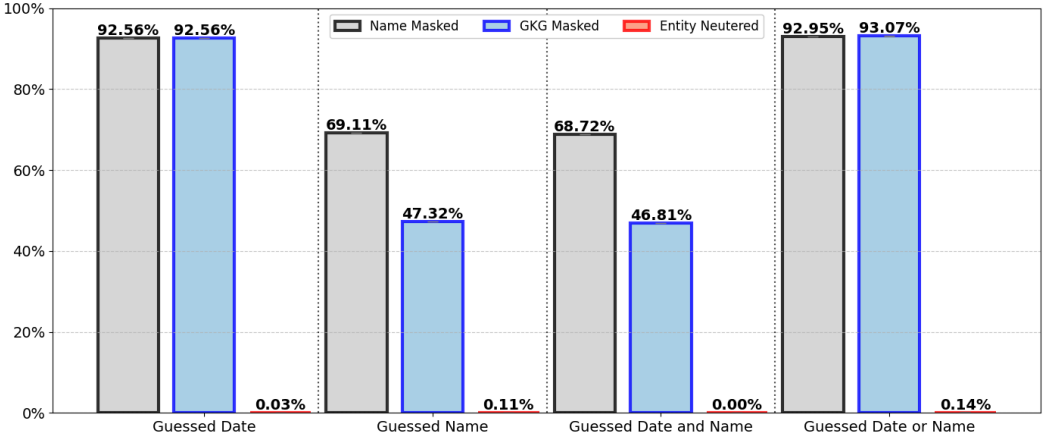
Neutering reduces firm recognition to about 0.12%

As good as random guessing after 9 iterations



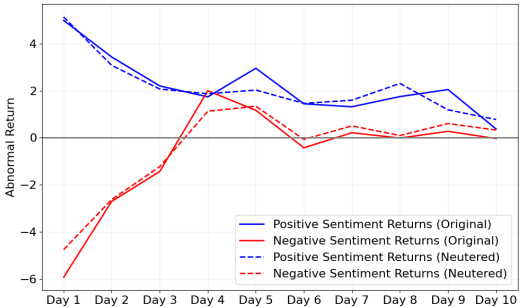
Entity neutering is considerably more effective than simpler masking

LLM guessing accuracy by de-identification method

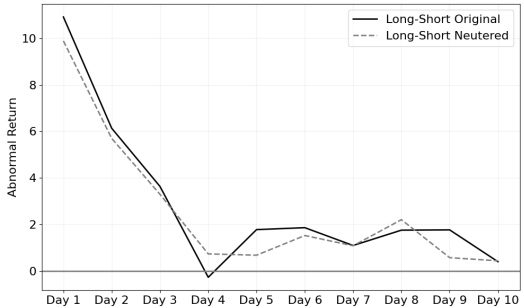


Sentiment from raw vs. neutered text yields similar return predictability

The difference provides an upper bound on lookahead bias



(a) Positive and negative



(b) Positive−negative

Entity neutering: Takeaways

- Entity neutering effectively anonymizes text and mitigates training leakage
- Semantic content is largely preserved
- Return predictability is nearly identical, bounding lookahead bias
- LLMs serve both as a source of bias and a tool for mitigation
- Python code at github.com/LukaVuli/Entity_Neutering

But are you sure there is no lookahead bias after neutering?

- If you need state-of-the-art LLMs, entity neutering is as good as it gets
- But it cannot guarantee that the LLM will not use information from the future
- If you absolutely want to be sure there is no lookahead bias, and don't require state-of-the-art LLMs, ChronoLLMs are for you

Conclusion: So which tool is right for your task?

1. **Chronologically Consistent LLMs**
   - If you need absolutely no lookahead bias
   - ChronoBERT and ChronoGPT are your workhorses
   - Provide clean baseline for lookahead bias measurement

2. **Instruction-Tuned ChronoLLMs**
   - If you need chat-capable models with chronological consistency
   - ChronoGPT-Instruct is your go-to
   - Best to fine-tune further for your application

3. **Entity Neutering**
   - If you need state-of-the-art LLMs and willing to tolerate some lookahead bias
   - Agentic approach to anonymize text to prevent LLM from using future knowledge
   - Quick, practical approach using SOTA existing LLMs